

Economic Policy Uncertainty, Public Attention, and DAX Volatility

Germany · January 2010 – January 2026 · Monthly data

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Research motivation and question

CORE QUESTION

Do periods of higher economic policy uncertainty coincide with stronger public search activity and higher DAX volatility?

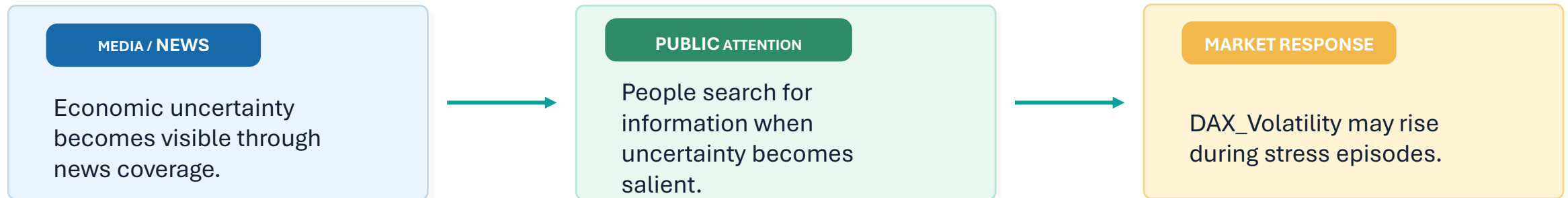
Uncertainty may make people look for more economic information. Google Trends offers a proxy for broad public attention. DAX_Volatility captures instability in the German stock market.

Focus: association, not causality



Media economics framing

The project is built on a media-information channel:



Media economics logic: the same information environment can both reflect uncertainty and shape attention.

Literature foundation

Media & finance**Tetlock (2015)**

Media can reflect and influence the financial information environment.

Measuring uncertainty**Baker, Bloom & Davis (2016)**

News-based text can be transformed into an Economic Policy Uncertainty index.

Attention channel**Barber & Odean (2008)**

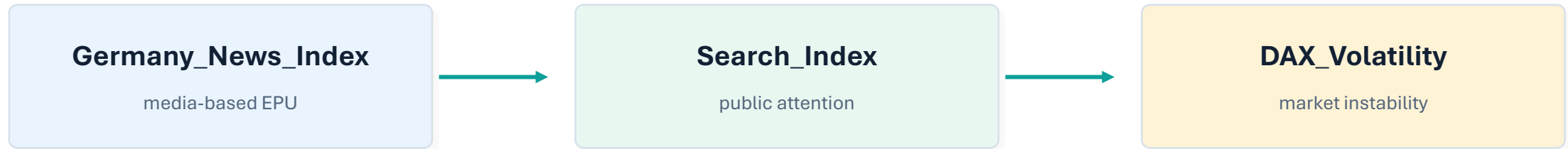
Individual investors are more likely to act on attention-grabbing information.

EPU and volatility**Liu & Zhang (2015)**

Policy uncertainty is linked to stock market volatility and financial stress.

Additional empirical motivation: Google Trends and search data can contain attention-related information for financial markets (Bock, 2018; Challet & Bel Hadj Ayed, 2014).

Project mechanism and hypotheses



Expected mechanism: uncertainty becomes visible in news, people search for information, and volatility may move with the same uncertainty environment.

- H1** Higher Germany_News_Index predicts stronger Search_Index.
- H2** Higher DAX_Volatility may increase subsequent Search_Index.
- H3** Search_Index may predict or Granger-cause DAX_Volatility.

Data overview

Germany

country focus

Jan 2010 – Jan 2026

monthly sample

193

observations

3 core variables

+ macro controls

Variable	Measurement	Role
Germany_News_Index	German news-based EPU index	Core variable: uncertainty
Search_Index	Average of 10 Google Trends keyword series	Core variable: public attention
DAX_Volatility	Monthly standard deviation of daily DAX returns	Core variable: market volatility
Controls	Inflation, unemployment, and ECB policy rate	Macro control variables

Search_Index is constructed from Google Trends keywords; controls are macro variables, not search keywords.

Variable construction

SEARCH_INDEX

$$Search_Index_t = \frac{1}{10} \sum_{k=1}^{10} GT_{k,t}$$

Search_Index = arithmetic average of ten Google Trends keyword series: inflation, recession, unemployment, interest rates, ECB, ECB interest rate, monetary policy ECB, Germany GDP, market crash, and banking crisis.

DAX RETURNS

$$r_d = \ln \left(\frac{P_d}{P_{d-1}} \right)$$

DAX_VOLATILITY

$$DAX_Volatility_m = \sqrt{\frac{1}{D_m - 1} \sum_{d \in m} (r_d - \bar{r}_m)^2}$$



Empirical strategy

METHOD

Vector Autoregression (VAR)

The VAR captures how Germany_News_Index, Search_Index and DAX_Volatility interact dynamically over time. Lag length $p = 3$ is used for short-run memory in attention and financial-market dynamics.

ORIGINAL VAR EQUATION

$$Y_t = Q_0 + \sum_{i=1}^3 Q_i Y_{t-i} + \varepsilon_t$$

where $Y_t = [\text{Germany_News_Index}_t, \text{Search_Index}_t, \text{DAX_Volatility}_t]'$

Structural shocks: $\varepsilon_t = A q_t$

VAR model

dynamic interdependence

IRF

shock propagation over 12 months

Granger tests

directional predictability

Cholesky ordering: Germany_News_Index → Search_Index → DAX_Volatility

Findings I: VAR dynamics

The VAR results show a dynamic relationship, but not a simple one-way chain.

- Germany_News_Index shows strong autoregressive and mean-reverting behavior.
- Lagged DAX_Volatility positively influences Germany_News_Index in the short run.
- Search_Index effects on Germany_News_Index are statistically weak or insignificant.
- DAX_Volatility is mostly internally driven and mean-reverting.

Next slide shows the original VAR output from the empirical analysis.

Original VAR output

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Summary of Regression Results
=====
Model: VAR
Method: OLS
Date: Tue, 23, Jun, 2026
Time: 04:30:21
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No. of Equations: 3.00000 BIC: 2.25227
Nobs: 189.000 HQIC: 1.94616
Log likelihood: -938.751 FPE: 5.68596
AIC: 1.73770 Det(Omega_mle): 4.87113
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Results for equation Germany_News_Index
=====
               coefficient      std. error      t-stat      prob
-----
const          8.096806          7.242413         1.118         0.264
L1.Germany_News_Index -0.475877          0.080542        -5.908         0.000
L1.Search_Index -2.109999          1.417456        -1.489         0.137
L1.DAX_Volatility 4146.350906        1691.226894         2.452         0.014
L2.Germany_News_Index -0.419046          0.083433        -5.023         0.000
L2.Search_Index  2.143990          1.489311         1.440         0.150
L2.DAX_Volatility 3206.212583        1792.486923         1.789         0.074
L3.Germany_News_Index -0.103833          0.080957        -1.283         0.200
L3.Search_Index -0.195977          2.094658        -0.094         0.925
L3.DAX_Volatility 2074.179264        1734.612768         1.196         0.232
=====

Results for equation Search_Index
=====
               coefficient      std. error      t-stat      prob
-----
const          0.575304          0.386934         1.487         0.137
L1.Germany_News_Index -0.011243          0.094303        -0.380         0.701
L1.Search_Index -0.030583          0.075729        -0.404         0.686
L1.DAX_Volatility  5.115256          90.355722         0.057         0.955
L2.Germany_News_Index -0.005533          0.004458        -1.241         0.214
L2.Search_Index -0.159468          0.079568        -2.004         0.045
L2.DAX_Volatility 46.422979          95.765654         0.485         0.628
L3.Germany_News_Index -0.018546          0.004325        -4.288         0.000
L3.Search_Index -0.239249          0.111909        -2.138         0.033
L3.DAX_Volatility 211.358913          92.673661         2.281         0.023
=====

Results for equation DAX_Volatility
=====
               coefficient      std. error      t-stat      prob
-----
const          0.000040          0.000357         0.112         0.911
L1.Germany_News_Index -0.000005          0.000004        -1.203         0.229
L1.Search_Index -0.000081          0.000070        -1.157         0.247
L1.DAX_Volatility -0.293282          0.083431        -3.515         0.000
L2.Germany_News_Index -0.000005          0.000004        -1.271         0.204
L2.Search_Index  0.000048          0.000073         0.649         0.516
L2.DAX_Volatility -0.148547          0.088427        -1.680         0.093
L3.Germany_News_Index 0.000000          0.000004         0.066         0.947
L3.Search_Index  0.000111          0.000103         1.077         0.282
L3.DAX_Volatility -0.021924          0.085572        -0.256         0.798
=====

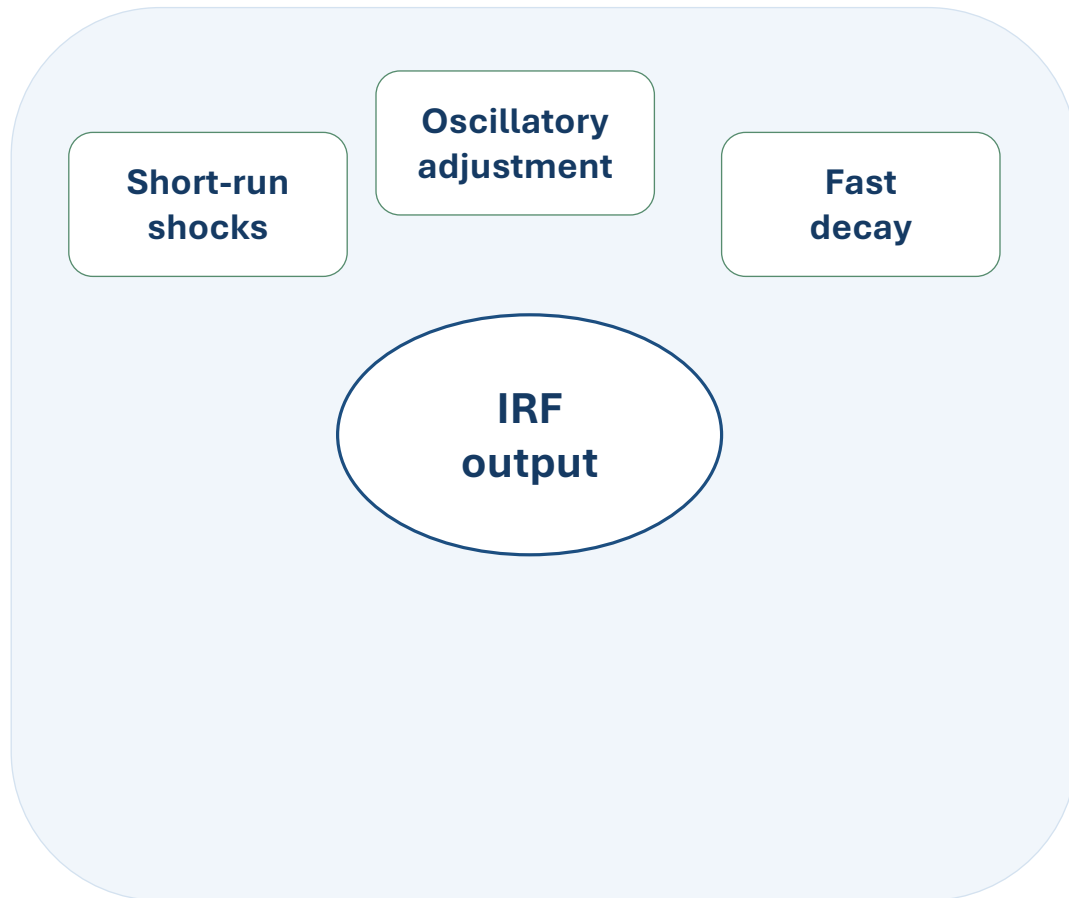
Correlation matrix of residuals
Germany_News_Index Search_Index DAX_Volatility
Germany_News_Index 1.000000 -0.028456 0.380926
Search_Index -0.028456 1.000000 0.262299
DAX_Volatility 0.380926 0.262299 1.000000

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Table 1: VAR analysis outcome

Original VAR output from the analysis; main interpretation is summarized in the surrounding slides.

Impulse responses: short-run reactions



Simplified IRF result: shocks are temporary and converge back toward zero.

Simplified impulse response interpretation

What the IRFs show

- Most responses are short-run and mean-reverting
- Germany_News_Index shocks mainly affect Search_Index
- Search_Index shocks do not create persistent DAX_Volatility responses
- The system converges back toward zero

Key message: temporary adjustment, not explosive dynamics.

Granger causality: Germany_News_Index → Search_Index

Relation	F-stat	p-value	df	Interpretation
Germany_News_Index → Search_Index	6.7540	0.000177	3	strong evidence
Search_Index → Germany_News_Index	1.4713	0.221399	3	not significant
Search_Index → DAX_Volatility	1.2003	0.308980	3	not significant
DAX_Volatility → Search_Index	1.7877	0.148428	3	not supported at 5%
Germany_News_Index → DAX_Volatility	0.9335	0.424205	3	not significant
DAX_Volatility → Germany_News_Index	2.4409	0.063486	3	marginal evidence

Main test results

Germany_News_Index → Search_Index is statistically significant at the 1% level.

Search_Index → DAX_Volatility is not significant.

DAX_Volatility → Search_Index is weak and not significant at the 5% level.

Result: the clearest evidence is from Germany_News_Index toward Search_Index.

Hypotheses: what was supported?

H1**Accepted**

Germany_News_Index predicts subsequent Search_Index.

H2**Weak evidence / not supported at 5%**

DAX_Volatility shows only weak evidence of predicting Search_Index, but the result is not statistically significant at the 5% level.

H3**Rejected**

Search_Index does not significantly predict DAX_Volatility.

Bottom line: Germany_News_Index predicts Search_Index, but Search_Index does not significantly predict DAX_Volatility.

Conclusion and limitations

MAIN TAKEAWAY

Media-based economic policy uncertainty is linked to public information search behavior, but Google search attention does not significantly predict DAX_Volatility.

Contribution

Connects Germany_News_Index, Search_Index and DAX_Volatility in a Germany monthly dataset using VAR, IRF and Granger tests.

Main limitations

No causality claim. Google Trends is a proxy. Keyword choices and major shocks may affect the stability of results.

Main result: Germany_News_Index predicts Search_Index; Search_Index does not robustly predict DAX_Volatility.